

Ricardo Costa

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ABOUT

Software engineering student focused on building reliable, scalable software. Main interests include **software verification**, **developer tools**, and **agentic AI workflows**. Currently working on LiquidJava, a static analysis tool that detects violations of properties specified through Java annotations at compile time, such as valid value ranges and API protocols. Also experienced in backend systems, web and mobile applications, and cloud computing.

EXPERIENCE

LASIGE Research Unit 🔗 2025 – 2026 (Expected)
Researcher 🔗 Lisbon, Portugal
Research: Improving the usability of LiquidJava, a liquid type system for Java, by enhancing error messages, extending IDE support, and refining the verification workflow to address usability barriers identified in an earlier paper 🔗
Contributions: 120+ merged pull requests to LiquidJava 🔗 and 80+ merged pull requests to its VS Code extension 🔗

EDUCATION

Faculdade de Ciências da Universidade de Lisboa (FCUL) 🔗 2024 – 2026 (Expected)
MSc in Computer Science and Engineering (Specialization in Software Engineering) Lisbon, Portugal
Current Grade: 17/20
Thesis: “Improving the Usability of LiquidJava” 🔗
Main Courses: Software Design (17/20), Compilation Techniques (18/20), Software Verification and Validation (16/20), Cloud Computing (18/20), Software Security (17/20), Mobile Computing (19/20)
Instituto Superior de Engenharia de Lisboa (ISEL) 🔗 2021 – 2024
BSc in Computer Science and Engineering Lisbon, Portugal
Final Grade: 15/20
Final Project: “NoteSpace 🔗: Multi-Platform Web Application for Real-Time Document Collaboration & Sharing” (18/20)
Main Courses: Programming (19/20), Software Laboratory (20/20), Algorithms and Data Structures (16/20), Web Programming (17/20), Mobile Programming (16/20)

PROJECTS

liquidjava 🔗 | Improved verification workflow and feedback with better error messages and developer experience
vscode-liquidjava 🔗 | Extended IDE support for reading, writing and debugging refinements in VS Code
liquidjava-mcp 🔗 | Java MCP server that exposes LiquidJava verification tools to LLM agents over stdio
NoteSpace 🔗 | Real-time collaborative editor with markdown and version control support
AI Code Reviewer 🔗 | Lightweight AI code review workflow powered by Cloudflare Workers AI and KV
Barista 🔗 | LLM agent workflow for synthesizing LiquidJava specifications from Javadoc documentation
Threadit 🔗 | REST API with microservices architecture, gRPC service communication, and GCP deployment
aguda-rs 🔗 | Compiler for a custom programming language implemented in Rust
passman 🔗 | Secure multi-user CLI password manager with local encrypted storage built in Rust
Sketch.io 🔗 | Multiplayer drawing and guessing game using React, Node.js, and Socket.IO
AdventOfCode 🔗 | Solutions for Advent of Code coding puzzles using Python (88 completed)

PUBLICATIONS

“Barista: Synthesizing Typestate Specifications with LLM Agents” | Catarina Gamboa, Paulo Canelas, [Ricardo Costa](#), Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Accepted at Automated Software Engineering (ASE) 2026 [📄](#))

POSTERS

“Improving the Usability of LiquidJava [📄](#)” | [Ricardo Costa](#), Catarina Gamboa, Alcides Fonseca

“Barista: Synthesizing Typestate Specifications with LLM Agents [📄](#)” | Catarina Gamboa, Paulo Canelas, [Ricardo Costa](#), Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Best PhD Student Poster Award)

TALKS

JNation 2026, Coimbra, Portugal [📄](#) | “LiquidJava: Catch Bugs at Compile Time with Liquid Types”

SKILLS

Programming Languages | Python, Java, JavaScript, TypeScript, Kotlin, Rust, Go, C, C#

Backend | REST, gRPC, Microservices, WebSockets, Spring Boot, Node.js, Express, Socket.IO, Maven, Gradle

Cloud & DevOps | GCP, Cloudflare Workers, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux

Frontend | React, HTML, CSS, Vite, Webpack, PWA

Mobile | Android, Jetpack Compose

Databases | PostgreSQL, Firestore, MongoDB, Cloudflare KV, Elasticsearch

Compilers & Static Analysis | Parsers, Type Systems, SMT Solvers, LLVM

Developer Tooling | VS Code Extension API, LSP, JSON-RPC

Testing | Unit Testing, Integration Testing, Property-Based Testing

AI Engineering | AI Agents, Agentic Workflows, Cloudflare Workers AI, MCP